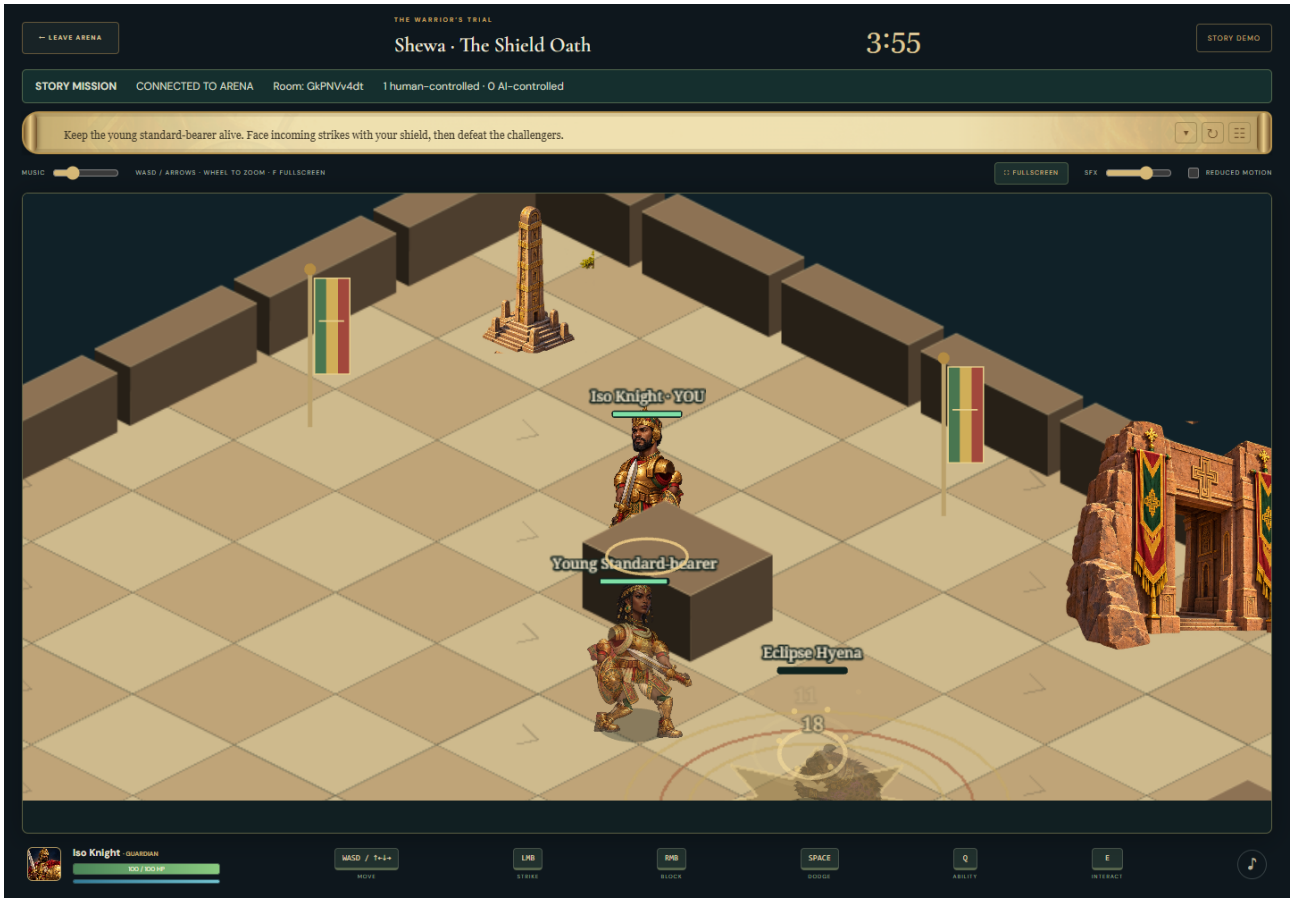
Kibre Gugs

THE BATTLE FOR THE THRONE · A REAL-TIME MULTIPLAYER ARENA



Live 2.5D isometric battle: painted hero heads, articulated warriors and the scroll guide.

What the game is

Kibre Gugs is a real-time multiplayer action game set in Aksumara, a fictional kingdom inspired by Ethiopian history and art. Players choose one of four heroes, fight live battles against other players and AI, and follow a 15-mission story campaign across three chapters. Every finished match produces a tamper-evident result record: the match, its winner, its timestamp and a cryptographic hash. Players can verify it and, when configured, it is attested on a public blockchain. A leaderboard, achievement badges and tournaments reward skill, and a real-world "lunch perk" rewards squads who meet in person.

The four heroes

- **The Lion Guard**: sword and shield; steady defence and strong strikes.
- **The Highland Spear**: long reach and quick thrusts.
- **The Rift Valley Archer**: ranged arrows from a distance.
- **The Royal Rider**: mounted fighter for riding and jousting missions.

Features

Game modes

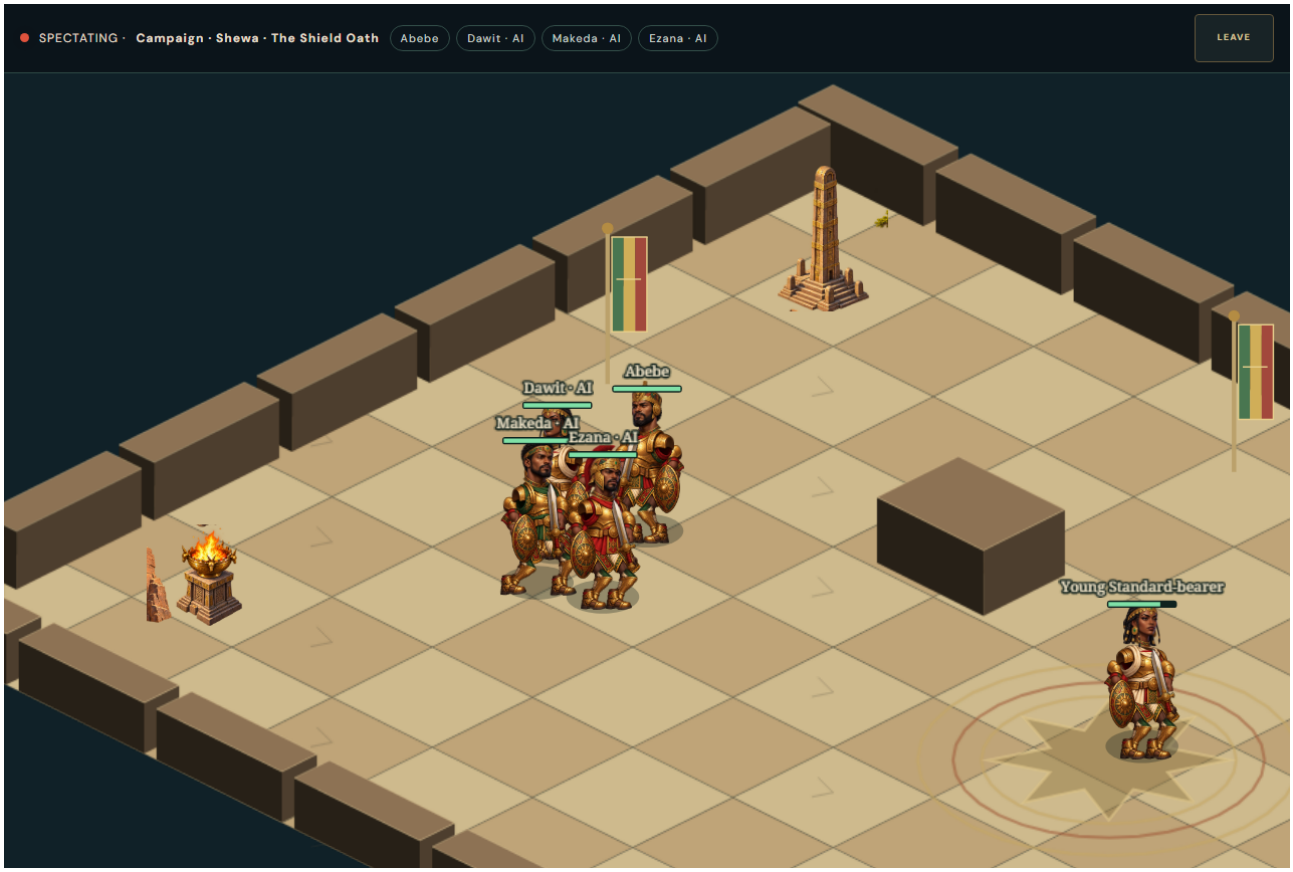
Mode	What happens
Training	A private tutorial that teaches movement, striking, blocking and dodging.
1v1 Duel	A live duel against another player; an AI opponent fills in after 5 seconds.
4v4 Banner Conquest	Two squads capture and hold banners across the arena.
Royal Quest	A four-player cooperative boss quest with puzzles.
Story campaign	15 missions in 3 chapters: protecting a standard-bearer, horse riding, a wind-swept joust with thrown spears, holding a ford against monster waves, rescues, escorts, puzzles, a guardian boss and the Eclipse Throne finale. A shorter 7-mission Story Demo is available for judges.
Tournament	A four-warrior knockout bracket (semifinals and final). Empty seats are filled by AI champions.

Gameplay

- An authoritative server simulates the whole battle 30 times per second; the browser predicts your own movement so controls feel instant.
- Combat with sword arcs, spear thrusts, arrows, a raised shield block, dodges with brief invulnerability, and a special ability per hero.
- Physics-driven animation: a running gait with knee lift, a wind-up / strike / follow-through sword swing, and spring-damped limbs, cape and weapon.
- Mounted jousting: aim with the cursor along a guide line; the thrown spear leaves the hand and flies as a real spear, bent slightly by the wind.
- Gradual difficulty: early missions send monsters one at a time with less health and weaker blows, rising to full strength by the final mission. Warriors recover between fights, and AI companions join from Chapter II.
- Restart after a loss and Next Stage after a win; progress is saved after every mission.

Multiplayer and social

- Live matchmaking: guests are matched with guests; identity-verified adults are matched within their own age band.
- Squads with six-character invite codes and QR passes, a ready-up lobby and a private, persistent team chat. The chat opens as a chat bubble and hides while a battle runs.
- Friends added by friend code, with online presence and squad invitations.
- Live notifications: a bell with an activity feed and pop-up toasts for friend requests, invites, saved results, badges, armor and tournament rounds.
- Spectator mode: a Live Matches list lets anyone watch a running battle and follow any fighter, without taking a seat.



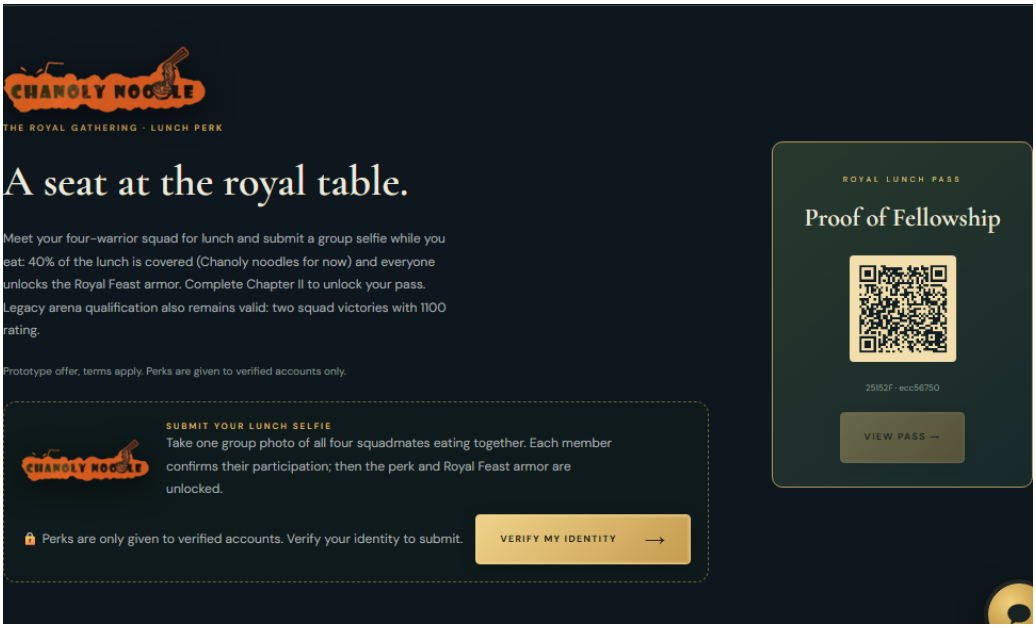
Spectator mode: watching a live campaign mission and choosing which fighter to follow.

Identity, verification and Web3

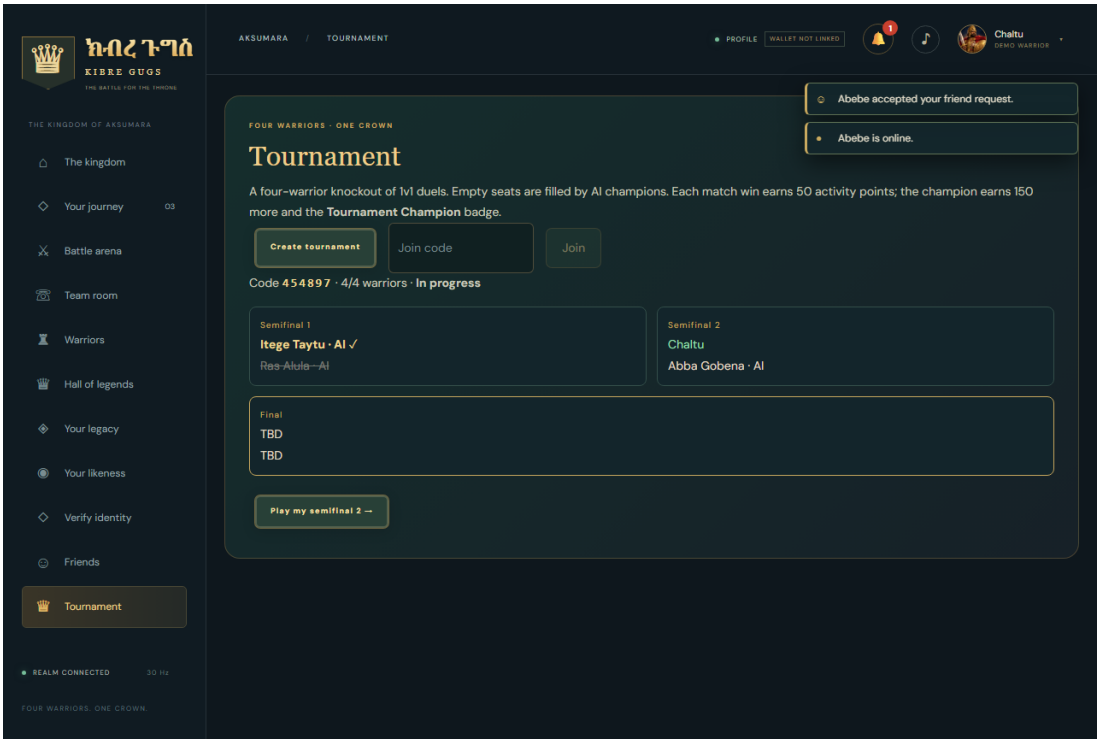
- **Match proofs:** each result is serialised canonically and hashed with Keccak-256. Players can re-check the hash in the browser from their Battle Chronicles.
- **On-chain attestation:** the RoyalRecords Solidity contract records match ID, winner, timestamp and result hash on Base Sepolia. Broadcasting is switched off by default and runs in clearly labelled simulation mode.
- **Wallet identity:** players can link a wallet by signing a single-use challenge; no custodial keys are created.
- **Achievements:** server-validated badges (First Blood, Sword Master, Thronebreaker, Tournament Champion and more), reward points and an ERC-1155 badge contract.
- **Leaderboard:** the Hall of Legends ranks players by rating, victories and XP.
- **Age verification:** players scan or upload a Fayda national ID. Text recognition runs on the device to suggest the birth date, a human reviewer approves it, and the player is notified of their eligibility.

Personalisation and rewards

- Painted hero heads by default, or a custom face built in the likeness editor. An optional camera capture suggests the face shape on the device; no selfie is uploaded.
- **Royal Lunch Perk** (prototype offer): when a four-warrior squad meets for lunch and submits a group selfie, the perk covers 40% of the lunch (Chanoly noodles for now) and unlocks the gilded **Royal Feast armor**. Perks are given to verified accounts only; players who decline keep playing.
- An original generated soundtrack with kebero war drums, a horn drone, a choir pad and a krar-style melody in the Ethiopian Tizita and Anchihoye modes.



The Royal Gathering lunch perk with the Chanoly logo and the selfie submission step.



Tournament bracket with AI-filled seats, and live notification toasts.

Tech stack

Layer	Technologies
Game client	React 19, Phaser 3.90 (2.5D isometric renderer), TypeScript 5.8, Vite 6, Zustand state, Tailwind CSS 4
Real-time networking	Colyseus 0.16 rooms over WebSockets (battles, campaign, squad lobbies); Socket.IO for team chat, notifications and spectators, sharing one port
Game server	Node.js 22, Express 4, zod input validation, a shared deterministic simulation package used by both server and browser
Database and auth	Supabase: PostgreSQL (private game_runtime schema with row-level security), email sign-in, private storage buckets and a scheduled Edge Function for evidence cleanup. SQLite is used for local development
Blockchain	Solidity 0.8.30 and OpenZeppelin 5.3 (RoyalRecords attestations and ERC-1155 badges), viem for signatures and Base Sepolia, Keccak-256 result hashing
On-device AI	MediaPipe Face Landmarker (face-shape suggestion) and Tesseract.js (ID birth-date OCR), both running locally in the browser
Audio	Web Audio API: a generated soundtrack and effects, with no audio files
Quality	Vitest unit and integration tests (including Postgres via PGlite), Playwright browser tests that play every mission, ESLint and Prettier
Hosting	A single Render Docker service serves the web app, API and WebSockets from one address; Supabase hosts data, auth and storage. pnpm monorepo

How it fits together

The browser loads the React app from Render and opens a WebSocket to the same service. Player inputs go to a Colyseus room, which runs the shared simulation at 30 Hz and streams snapshots back. When a match ends, the server writes the result, its hash, progress and badges to Supabase Postgres in one transaction, queues the optional blockchain attestation, and sends a live notification. Chat, notifications and spectator feeds travel over Socket.IO on the same port.

Honest limitations

- Blockchain attestation runs in simulation unless the contract and attestor key are configured; simulated records are never presented as on-chain proof.
- ID review is a manual age check by a reviewer, not government authentication.
- The lunch perk is a prototype offer; the game does not process payments.
- On free hosting tiers the server sleeps when idle and takes about a minute to wake.